Exercise 1: Inventory Management System

Aim

To implement an Inventory Management System using Java and understand how data structures help in storing and managing product information efficiently.

Problem Understanding

Managing a warehouse inventory involves storing and retrieving product details quickly. Data structures and algorithms make operations like adding, updating, searching, and deleting products more efficient, especially when the number of products is large.

Data Structure Used

HashMap<ProductId, Product>

HashMap is used because it provides fast access to product information using the product ID as the key.

Classes Used

Product

Represents a product with the following attributes:

- Product ID
- Product Name
- Quantity
- Price

InventoryManager

Manages the inventory by performing operations such as:

- Add Product
- Update Product
- Delete Product
- Display Products

Main

Contains the main method used to test the inventory management operations.

Operations Performed

- Add new products to the inventory
- Update existing product details
- Delete products from the inventory
- Display all available products

Time Complexity

Operation	Complexity
Add Product	O(1) (Average)
Update Product	O(1) (Average)
Delete Product	O(1) (Average)
Search Product	O(1) (Average)

Optimization

Using HashMap avoids searching through every product one by one, making the inventory operations faster and more suitable for large datasets.

Output

```
Product Added Successfully.
Product Added Successfully.

Inventory Details
--------------------------
Product ID    : 101
Product Name  : Laptop
Quantity      : 20
Price         : 55000.0

Product ID    : 102
Product Name  : Mouse
Quantity      : 50
Price         : 700.0

Product Updated Successfully.

Inventory Details
--------------------------
Product ID    : 101
Product Name  : Laptop
Quantity      : 20
Price         : 55000.0

Product ID    : 102
Product Name  : Wireless Mouse
Quantity      : 45
Price         : 900.0
```

```
Product Deleted Successfully.

Inventory Details
--------------------------
Product ID   : 102
Product Name : Wireless Mouse
Quantity     : 45
Price        : 900.0



...Program finished with exit code 0
Press ENTER to exit console.
```

Conclusion

This exercise demonstrates the use of HashMap for efficient inventory management and shows how appropriate data structures improve the performance of CRUD operations in Java applications.